

Fluid Flow in Pipes

Final Lab Report
Unit Operations Lab 4
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1. Executive Summary

The objective of this experiment was to quantify energy losses in a pipe system by determining the flow coefficient (C_f) of an orifice meter, the friction factor (f) for straight pipes, and the loss coefficient (K) for various fittings. These parameters are essential for predicting head losses, selecting pumps, and designing efficient fluid transport systems.

Flowrates, differential pressures, and pipe dimensions were measured for several pipe sections and fittings under different operating conditions. Each geometry of interest received three different flowrates. Calibration of the flowmeter established a conversion factor of $0.8978 \text{ L min}^{-1}$ per SCFM, confirming the accuracy of the scale used. The measured flow coefficients ranged from 0.673 to 0.694, showing little dependence on flowrate and aligning with expected values of 0.6–0.9. Friction factors for straight pipes decreased from approximately 0.042 to 0.018 as the Reynolds number increased, consistent with the transition from laminar to turbulent flow. Calculated head losses ranged between 0.2 and 7 m, matching literature trends, while fitting loss coefficients increased with velocity, with elbows producing the largest local losses.

Overall, the experimental data followed established fluid-flow correlations within reasonable uncertainty. Variations were mainly due to small pressure drops near the lower sensitivity limit of the gauges. The findings confirm that pressure and head losses increase with flow velocity and depend strongly on geometry, supporting theoretical models used in process piping design and reinforcing the need for accurate loss coefficients in practical engineering calculations.

2. Introduction

The main objective of this experiment is to identify energy losses in pipe systems through obtaining different fluid flow parameters like the friction factor (f), the loss coefficient (K), and the flow coefficient for different fittings pipe sections, and an orifice meter in the lab. The values obtained will be compared against theoretical models and correlations for the turbulent and laminar flow.

Pressure drops and head loss are important in chemical engineering practices as they are crucial in industrial processes. Fluids moving between reactors, heat exchangers, storage systems, and separation units are all processes that involve understanding pressure drop and head loss. It is important that engineers can predict pressure loss due to fittings and pipe friction as it's needed to size and select pumps and design piping layouts that allow for the efficient movement of flowrates. In multiple plants, it's common that the systems are not only single-phase liquid and gas flow but multiphase mixtures and slurries where empirical correlations are a requirement when the theory is not enough.

The apparatus involved in this experiment is a Hampden H-6920 Pipe Friction Demonstrator, which is a flow loop equipped with various replaceable pipe sections, fittings, orifice flow meter, recirculating water pump and reservoir. The system also contains differential pressure transmitters (DPTs), manual pressure gauges, and a rotameter for flow. These components allow for measurement of pressure drops through the fittings, pipe lengths, and the orifice plate. Also measured are the fluid temperature and volumetric flowrate. Through these measurements, we can see the small losses in fittings, the head loss in straight pipes and the flow coefficient of the orifice meter can be gotten as functions of the Reynolds number.

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Industrial applications of this work specifically include line sizing, pump specification, and correcting energy losses in process piping. Typically, engineers use K values, friction factors, and meter coefficients to model networks, project operating costs, and make sure that systems operate in their safe pressure limits. This lab is relevant to real-world plant design and operations.

3. Theory

The fluid flow in pipe experiment is designed to study energy losses in pipe systems through theories and empirical correlations.

Orifice Meter Flow Measurement

An orifice flowmeter is used to measure the volumetric flowrate through relations with the pressure differential across the orifice plate. The volumetric flow rate formula for real, incompressible flows is:

$$\dot{Q} = C_f A_o \sqrt{\frac{2\Delta P}{\rho}} \text{ (Eq1)}$$

Where C_f is the orifice coefficient which accounts for viscosity and turbulence. A_o is the area of the orifice; ΔP is the differential pressure ($P_1 - P_2$) across the orifice and ρ is the fluid density. The orifice coefficient is often associated with the ratio of the orifice diameter. (Fluid Flow Lab Handout Page 3-5)

Pipe Friction Head Loss

The energy loss due to friction in fully developed flow in straight pipes is expressed as head loss which is proportional to the friction factor. The equation is

$$h_L = \frac{\Delta P}{\rho} = f \frac{L}{D} \frac{V^2}{2} \text{ (Eq2)}$$

Where ΔP is the pressure drop over the straight length, ρ is the fluid density, L is the pipe length, D is the inner diameter, and V is the average fluid velocity. You can also find the friction factor (f) through an experiment using the relation

$$f = \frac{\Delta P D}{2 L \rho V^2} \text{ (EQ3)}$$

The resistance in laminar flow is because of viscous forces, and the friction factor is given as:

$$f = \frac{64}{Re} \text{ (EQ4)}$$

The friction factor for turbulent flow is dependent on the Reynolds number and the roughness of the conduit walls k_s/D (Fluid Flow Lab Handout Page 5).

Pipe Fitting Head Loss

Fittings like elbows and bends, introduce new energy losses due to increased turbulence and flow separation. The head loss due to pipe fitting can be estimated using the loss coefficient (K)

$$h_L = K \frac{V^2}{2} \text{ (Eq5)}$$

(Fluid Flow Lab Handout Page 6).

The loss coefficient is specific to the fitting type and the head loss can be express through an equivalent length of diameter (L_{eq}/D) by relating the K to the friction factor by:

$$K = f \frac{L_{eq}}{D} \text{ (EQ6)}$$

(Fluid Flow Lab Handout Page 6).

4. Results

The key measurements made in lab utilized in calculations of the flow coefficient, head loss and pipe fitting head loss, were the pressure drop, length of the pipe, volumetric flow rate, and pipe diameter. To perform the experiment the flowrate of SCFM of the air must have been converted to a volumetric flow rate of water. This was accomplished by setting various flow rates of SCFM and recording the volume of water collected in a bucket for a given time. The time was standardized to 5 seconds, and the result is the following graph:

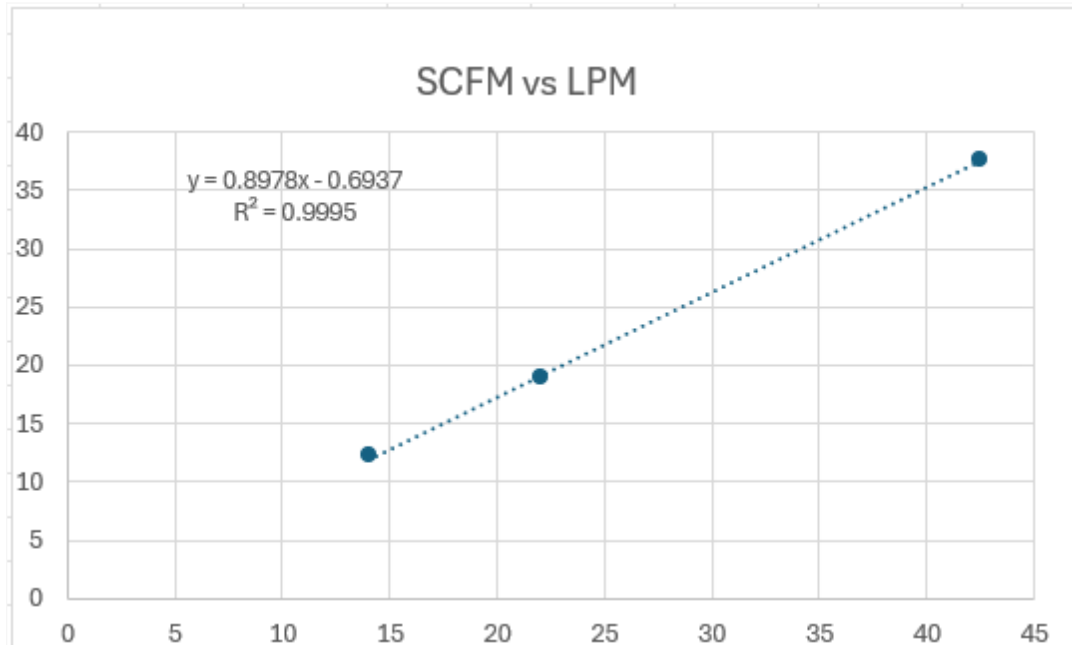


Figure 1 SCFM vs Volumetric Flow rate

The slope of the graph was the conversion factor between SCFM and liters per minute, which ended up being 0.8978. This factor was used to calculate the volumetric flow rate which was in turn used to calculate the velocity:

$$v = \frac{Q}{A} \quad (Eq 7)$$

An online calculator was utilized to calculate both the density and viscosity of water which were utilized in the calculations of the Reynold's number and the friction factor^[1].

To ensure consistent results a value of the pressure was recorded every 5 seconds for 20 seconds and an average was taken. Since the pressure drop is measured in psi, it must also be converted to pascals which is accomplished by the following formula:

$$P_{Pa} = P_{psi} \times 6894.76 \quad (Eq 8)$$

With the calculated Pressure, velocity, velocity the flow coefficients are calculated utilizing the formula:

$$C_f = \frac{Q}{A \sqrt{2 \Delta P / \rho}} \quad (Eq 9)$$

The tabulated flow coefficients are as shown in figure 2:

SCFM	CF
45	0.673
40	0.688
30	0.672
19	0.694

Figure 2 SCFM vs Flow coefficients

The values for the flow coefficient ranged between .673 and .694, not following any clear trends. This suggests that the flow coefficient does not vary with varying flow rates since no clear trend was observed. The variation likely comes from experimental errors likely in instrumentation. The recorded pressure drops are on the lower side of the range for the pressure gauge, which leads to uncertainty in the measurements leading to

uncertainty in the pressure values which directly effects the calculation of the flow

coefficient C_f . This value agrees with the value range given in the lab manual of 0.6-0.9

The calculation for the head loss requires many of the same quantities along with the pipe geometry. The first value that must be calculated is the friction factor using the equation:

$$f = \frac{\Delta P D}{2 L \rho V^2} \quad (EQ 3)$$

The pipes had uniform length and only varied in their diameter. A table of the tabulated friction factor using the diameters is as shown in figure 3:

Friction factor	Small1	Small 2	small 3	Mid1	Mid2	Mid3	Large1	Large 2	Large 3
45 SCFM							0.018	0.012	0.035
40 SCFM							0.526	0.259	0.526
30 SCFM							0.033	0.016	0.033
19 SCFM	0.032	0.020	0.083	0.049	0.017	0.051	0.027	0.018	0.033
15 SCFM	0.035	0.024	0.096	0.037	0.022	0.060			
10 SCFM	0.045	0.029	0.116	0.052	0.027	0.072	0.000	0.000	0.000

Figure 3 Friction Factors vs SCFM

As the flowrate decreased the friction factor increased which follows the trends

described in equation 4. The values ranged between what is expected in literature

With the calculated friction factor the head loss can be calculated using the equation:

$$h_L = \frac{\Delta P}{\rho} = f \frac{L}{D} \frac{V^2}{2} \quad (EQ 5)$$

The results are as follows in figure 4:

Head Loss	Small1	Small 2	small 3	Mid1	Mid2	Mid3	Large1	Large 2	Large 3
45 SCFM	0.000	0.000	0.000	0.000	0.000	0.000	0.326	0.368	0.017
40 SCFM	0.000	0.000	0.000	0.000	0.000	0.000	7.422	6.439	0.079
30 SCFM	0.000	0.000	0.000	0.000	0.000	0.000	0.261	0.227	0.277
19 SCFM	1.297	2.913	3.355	0.722	0.916	0.750	0.046	0.063	0.055
15 SCFM	0.882	2.112	2.421	0.340	0.732	0.552	0.000	0.000	0.000
10 SCFM	0.502	1.156	1.297	0.212	0.403	0.297	0.000	0.000	0.000

Figure 4 Head loss vs SCFM

Any zeros do not indicate zero head loss or zero friction instead they indicate that no data

was available at that flow rate. For example, the flow rate was unable to reach 45 SCFM

utilizing the small flow valves, so the values of head loss of those are unknown for the

smaller and middle-sized pipes. Values ranged from .2 to 7 which are in range for literature values, values in literature ranged from .3 at .01 bar drop to 20 at 1 bar^[4]. To calculate the values of the Pipe fitting head loss the loss coefficient K must be calculated using the equation:

$$K = f \frac{L}{D} \quad (EQ 6)$$

The friction factors were calculated once again for the pipe fittings, and the corresponding lengths and diameters were input to give the following values for the K values:

K VAL			Exp(1)	Contractio	Elbow	Oraface	Bend	Utube
40 SCFM			0.121	0.048	0.116	0.018	0.021	0.050
30 SCFM			0.153	0.049	0.116	0.017	0.016	0.069
19 SCFM			0.163	0.061	0.137	0.018	0.023	0.075
15 SCFM			0.407	0.051	0.274	0.017	0.058	0.117
10 SCFM			0.653	0.081	0.384	0.026	0.111	0.225

Figure 5 Tabulated K VAL

Utilizing these values to calculate the pipe fitting head loss using the equation:

$$h_L = K \frac{V^2}{2} \quad (EQ 7)$$

Results in the following values for the pipe fitting head loss(hl):

Pipe fitting head loss		Exp(1)	Contractio Elbow	Oraface	Bend	Utube	
40 SCFM		0.002	0.005	0.007	0.001	0.001	0.002
30 SCFM		0.002	0.004	0.005	0.001	0.001	0.002
19 SCFM		0.001	0.003	0.003	0.000	0.001	0.001
15 SCFM		0.001	0.001	0.003	0.000	0.001	0.001
10 SCFM		0.001	0.001	0.002	0.000	0.001	0.001

Figure 6 Pipe fitting head loss VS SCFM

For all a given pipe fitting increasing the flowrate and therefore the velocity increased the value of Hl. This tracks with the relationship in equation 7 as they are directly correlated. The largest changes were seen in the elbow which was unexpected as the

pressure drop was the largest in the orifice. With a larger pressure drop a larger friction factor would be expected leading to a larger Hl value but this is also affected by geometric features that appear to have offset the difference in pressure. The Reynolds number is calculated using the equation:

$$Re = \frac{\rho V D}{\mu} \quad (EQ 8)$$

This was calculated for all the pipe fittings and pipe geometries. The graph of Reynolds number vs friction factor for a single pipe was as follows:

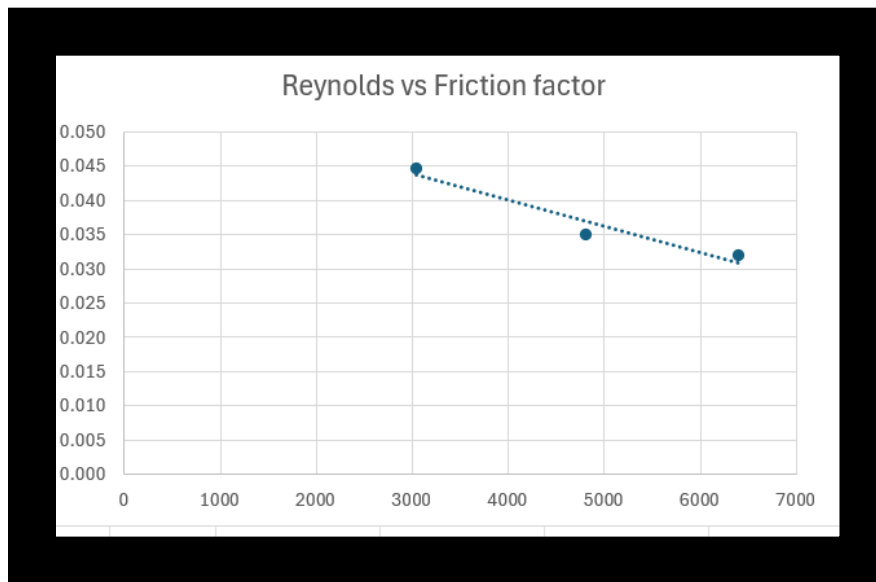


Figure 7 Reynolds vs Friction Factor

This trend was followed in all trials with the friction factor decreasing as the Reynolds number increases. The graph for pipe Hl vs Reynolds number is as shown in figure 8:

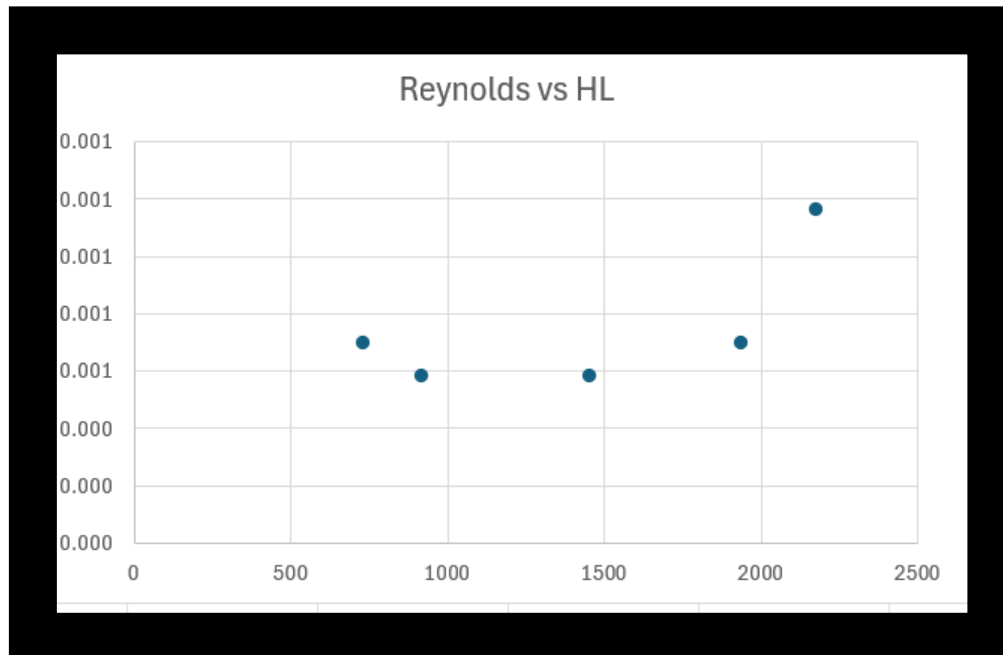


Figure 8 HL vs Reynolds #

As the Reynolds number increases HL also increased albeit nonlinearly. This is consistent with equation 7 as with a higher velocity the KH is also larger.

5. Discussion

The objective of this experiment was to evaluate the head losses and loss coefficients (K) associated with different pipe fittings under varying air flow rates and to compare the results with theoretical and literature values. The fittings tested included contractions, elbows, orifices, and bends, with flowrates ranging from 10 to 45 SCFM. The experiment also involved determining the discharge coefficient (Cf) of an orifice meter and the friction factor (f) for straight pipes to assess energy losses in the system.

The results demonstrated strong agreement with established fluid mechanics theory. Head loss and measured pressure drop increased with air velocity, consistent with the relationship

$$h_L = \frac{\Delta P}{\rho} = f \frac{L}{D} \frac{V^2}{2} \quad (EQ 5)$$

The orifice produced the largest local losses, with $K \approx 0.65$ at 10 SCFM and remaining above 0.1 at higher velocities. In contrast, the bend and contraction fittings exhibited moderate losses ($K \approx 0.01\text{--}0.05$), and the elbow showed the smallest resistance. These trends align with standard references such as Geankoplis (2018), which report that sharp restrictions and orifices create greater turbulence and flow separation than smooth bends or elbows. The measured discharge coefficients ranged from 0.673 to 0.694, which falls within the expected literature range of 0.6–0.9. The friction factors decreased from approximately 0.042 to 0.018 as the Reynolds number increased, reflecting the expected shift from laminar toward turbulent flow.

Overall, the data are consistent with the work of others and with standard models for pipe flow and minor losses. Minor deviations from theoretical values likely result from measurement uncertainty and non-ideal experimental conditions. For example, small differential pressures at low flowrates approached the resolution limit of the U-tube manometer, introducing relatively large percentage errors. In addition, slight flow instability in the rotameter and potential leaks at the pressure taps may have caused underestimation of the true pressure losses. Air compressibility at higher flowrates and assumptions of uniform velocity across the pipe cross-section may have introduced further errors. These limitations, however, did not obscure the key physical relationships between flowrate, head loss, and fitting geometry.

The experimental method was effective in capturing the overall behavior of head loss across different fittings, but it could be improved by replacing the U-tube manometer with a digital differential pressure transducer to improve resolution. Ensuring all joints

and taps are fully sealed and using a flowmeter with greater precision would further reduce uncertainty. Additionally, collecting more data points at mid-range flowrates could help smooth out measurement scatter.

The assumptions made; steady-state flow, negligible compressibility, and uniform cross-sectional velocity. These were generally reliable under the conditions tested. However, compressibility effects may become more relevant at the highest flowrates, and the steady-flow assumption could be weakened by slight discharge fluctuations during longer trials.

Definite conclusions from this experiment are that pressure drop, and head loss increase with fluid flowrate, orifices create the largest local losses, friction factor decreases with Reynolds number, and discharge coefficients remain nearly constant within the expected range. More conclusions include precise quantitative agreement with specific literature K-values, which depend on detailed geometry and flow uniformity.

These findings have direct implications for engineering design and practice. They confirm that sharp restrictions in piping system such as orifices and contractions can significantly increase energy losses and must be considered during pump selection and process design. The relative ranking of K-values also validates the use of standard loss coefficient tables for preliminary system design.

6. References

[1]The Engineering ToolBox. (2004). *Water – Dynamic and kinematic viscosity at various temperatures and pressures*. Retrieved October 27, 2025, from https://www.engineeringtoolbox.com/water-dynamic-kinematic-viscosity-d_596.html [1]

[2]The Engineering ToolBox. (n.d.). *Water – Density, specific weight and thermal expansion coefficients*. Retrieved October 27, 2025, from

[3]https://www.engineeringtoolbox.com/water-density-specific-weight-d_595.html

Friction Factor. In *ScienceDirect Topics*. Elsevier. Retrieved October 27, 2025, from

[4]<https://www.sciencedirect.com/topics/engineering/friction-factor>[2]

Crane Co., NY. (1982) *Flow of Fluids through Valves, Fittings and Pipes*. Metric Edition-SI Units, Technical Paper No. 410M, Crane Co., New York.

7. Appendix I: Data Tabulation/Graphs

Flowrate(S	Flowrate(w	Water temp	Exp(1)	Contractio	Elbow	Orifice	Bend	Utube	Small1	Small 2	small 3	Mid1	Mid2	Mid3	Large1	Large 2	Large 3
40	35.9148	73.1	0.1	0.13	0.15	3.86	0.06	0.13		0.2			0.05	0.1			
Large pipes																	
19	17.0595	76.2	0.06	0.03	0.08	0.86	0.05	0.05		0.75	1.31	1.94	0.51	0.5	0.53	0.05	0.06
t=5			0.06	0.03	0.08	0.86	0.05	0.06		0.75	1.3	1.93	0.52	0.5	0.53	0.05	0.06
	10		0.05	0.03	0.07	0.85	0.05	0.05		0.76	1.31	1.9	0.35	0.5	0.53	0.05	0.06
	15	78.2	0.06	0.03	0.07	0.85	0.06	0.06		0.75	1.31	1.91	0.35	0.5	0.53	0.05	0.07
30	26.9361	78.5	0.06	0.09	0.1	2.27	0.05	0.08								0.07	0.09
			0.05	0.09	0.11	2.28	0.06	0.08								0.08	0.1
			0.06	0.09	0.11	2.26	0.06	0.09								0.07	0.09
			0.07	0.09	0.1	2.28	0.06	0.09								0.08	0.09
15	13.4681	79.3								0.51	0.95	1.4	0.24	0.4	0.39		
										0.53	0.94	1.4	0.25	0.4	0.37 CF		
										0.53	0.95	1.39	0.25	0.4	0.38		
										0.53	0.95	1.39	0.24	0.4	0.37		
45	40.4042	79.7	0.1	0.16	0.19	5.08	0.1	0.12									
			0.11	0.15	0.19	5.06	0.1	0.13									
			0.1	0.15	0.19	5.11	0.09	0.13									
			0.1	0.16	0.2	5.14	0.1	0.13									
										0.3	0.51	0.75	0.15	0.23	0.22		
										0.28	0.52	0.75	0.13	0.21	0.21		
										0.29	0.52	0.75	0.15	0.22	0.21		
10	8.9787	80.4								0.28	0.51	0.75	0.15	0.21	0.21		

Figure 9 RAW DATA

viscosity c	density of v	VOL (m ³ /3/	ΔP pas	Q/Ao	sqrt(rho/2	Cf
0.9346	998	0.00060	26614	5.017	0.137	0.688
0.8981	997	0.00028	5895	2.383	0.291	0.694
	997					
0.8758	997	0.00045	15667	3.763	0.179	0.672
0.8639	997	0.00022		1.882		
0.8596	997	0.00067	35146	5.645	0.119	0.673
0.8523	997	0.00015				

Figure 10 Data for calculation of Cf

[illegible]

Figure 11 Measured and calculated system geometry

		Exp(1)	Contractor		Elbow	Orifice	Bend	Utube	Small1	Small2	small3	Mid1	Mid2	Mid3	Large1	Large2	Large3
Velocity	45 SFM		0.17	0.45	0.34	0.34	0.34	0.26	2.59	4.29	2.59	1.74	2.91	1.74	0.73	0.97	0.73
	40 SCFM		0.15	0.40	0.30	0.30	0.30	0.23	2.30	3.81	2.30	1.54	2.58	1.54	0.65	0.86	0.65
	30 SCFM		0.11	0.30	0.22	0.22	0.22	0.17	1.73	2.86	1.73	1.16	1.94	1.16	0.48	0.64	0.48
	19 SCFM		0.07	0.19	0.14	0.14	0.14	0.11	1.09	1.81	1.09	0.73	1.23	0.73	0.31	0.41	0.31
	15 SCFM		0.06	0.15	0.11	0.11	0.11	0.09	0.86	1.43	0.86	0.58	0.97	0.58	0.24	0.32	0.24
	10 SCFM		0.04	0.10	0.07	0.07	0.07	0.06	0.58	0.95	0.58	0.39	0.65	0.39	0.16	0.21	0.16
Pressure(PA)		Exp(1)	Contractor		Elbow	Orifice	Bend	Utube									
	45 SCFM		689	1103	1310	35025	689	827									
	40 scfm		689	896	1034	26614	414	896							689	896	1310
	30 SCFM		414	621	689	15651	345	552							552	689	827
	19 SCFM		414	207	552	5929	345	345	5171	9032	13376	3516	3447	3654	345	414	414
	15 SCFM		414	207	483	5861	414	414	3516	6550	9653	1655	2758	2689			
	10 SCFM		345	207	483	5861	345	345	1999	3585	5171	1034	1517	1448			

Friction factor	Exp(1)	Contractio	Elbow	Orifice	Bend	Utube	Small1	Small2	Small3	Mid1	Mid2	Mid3	Large1	Large2	Large3
40 SCFM	0.058	0.021	0.043	0.012	0.014	0.029	0.022	0.000	0.006	0.022	0.060	0.016	0.060	0.526	0.035
30 SCFM	0.073	0.021	0.043	0.011	0.011	0.040								0.526	0.035
19 SCFM	0.078	0.026	0.051	0.012	0.016	0.044	0.032	0.020	0.083	0.049	0.017	0.051	0.027	0.018	0.033
15 SCFM	0.194	0.022	0.103	0.011	0.040	0.068	0.035	0.024	0.096	0.037	0.022	0.060			
10 SCFM	0.311	0.035	0.144	0.017	0.078	0.131	0.045	0.029	0.116	0.052	0.027	0.072	0.000	0.000	0.000

Figure 13 Friction Factors

K VAL	Exp(1)	Contractio	Elbow	Orifice	Bend	Utube	Small1	Small2	Small3	Mid1	Mid2	Mid3	Large1	Large2	Large3
40 SCFM	0.121	0.043	0.116	0.018	0.021	0.050	0.000	0.000	0.000	0.000	0.000	0.000	0.326	0.368	0.017
30 SCFM	0.153	0.049	0.116	0.017	0.016	0.069	0.000	0.000	0.000	0.000	0.000	0.000	0.261	0.227	0.277
19 SCFM	0.163	0.061	0.137	0.018	0.023	0.075	1.297	2.913	3.355	0.722	0.916	0.750	0.046	0.063	0.055
15 SCFM	0.407	0.051	0.274	0.017	0.058	0.117	0.882	2.112	2.421	0.340	0.732	0.552	0.000	0.000	0.000
10 SCFM	0.653	0.081	0.384	0.026	0.111	0.225	0.502	1.156	1.297	0.212	0.403	0.297	0.000	0.000	0.000
Pipe fitting head loss	Exp(1)	Contractio	Elbow	Orifice	Bend	Utube	Small1	Small2	Small3	Mid1	Mid2	Mid3	Large1	Large2	Large3
40 SCFM	0.002	0.005	0.007	0.001	0.001	0.002	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
30 SCFM	0.002	0.004	0.005	0.001	0.001	0.002	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
19 SCFM	0.001	0.003	0.003	0.000	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
15 SCFM	0.001	0.001	0.003	0.000	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
10 SCFM	0.001	0.001	0.002	0.000	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Head Loss	0.0121	0.125													

Figure 14 K values and Head losses

REYNOLDS	Exp(1)	Contractio	Elbow	Orifice	Bend	Utube	Small1	Small2	Small3	Mid1	Mid2	Mid3	Large1	Large2	Large3
45 SCFM	668	1782	1337	1337	2180	1677	7200	9257	7200	5891	9866	5891	3812	4393	3812
40 SCFM	594	1584	1188	1188	1937	1490	6400	8228	6400	5236	8769	5236	3388	3905	3388
30 SCFM	446	1188	891	891	1453	1118	4800	6171	4800	3927	6577	3927	2541	2929	2541
19 SCFM	282	752	564	564	920	708	3040	3909	3040	2487	4166	2487	1609	1855	1609
15 SCFM	223	594	446	446	727	559	2400	3086	2400	1964	3289	1964	1271	1464	1271
10 SCFM	149	396	297	297	484	373	1600	2057	1600	1309	2192	1309	847	976	847

Figure 15 Reynold's Numbers

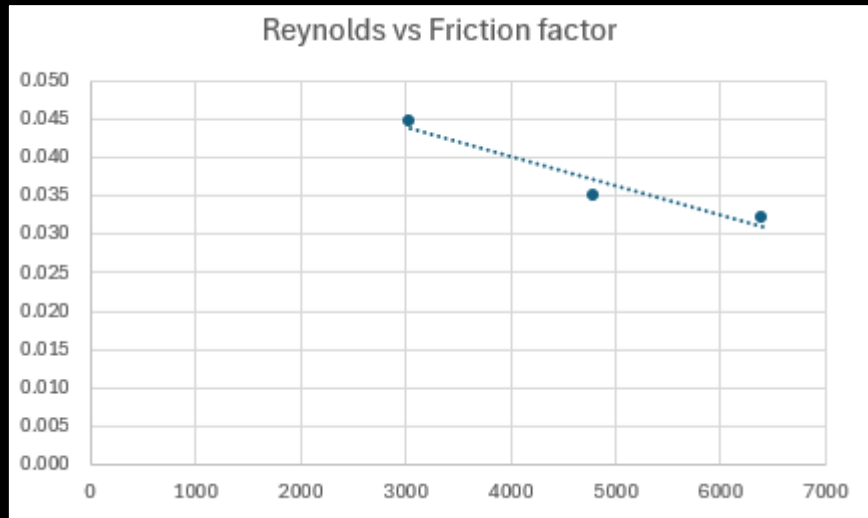


Figure 16 Friction Factor vs Reynolds for a selected pipe

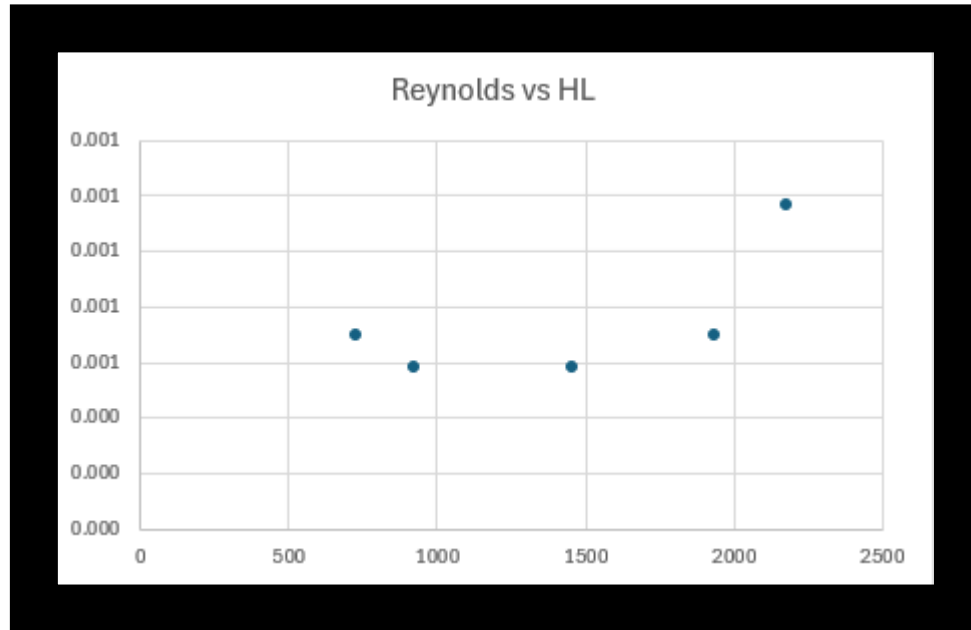


Figure 17 HL vs Reynolds Number

8. Appendix II: Error Analysis

The error analysis takes into consideration the experimental errors during the analysis of the fluid flow data. Various sources determine the qualitative errors. The fluctuations in the DPT readings show variety in the values of the measured pressure drop which increases the uncertainty in the calculated coefficients. Another issue faced were air bubbles in the pressure lines and the sensing ports which resulted in irrelevant pressure readings however this can be corrected by following the procedure to flush out the DPT lines. As the fluid properties are dependent on temperature, any change in water temperature can affect the density and fluid viscosity which influences the Reynolds number and other calculated dimensionless coefficients. These calculated coefficients could display a slight difference from empirical correlations or literature values due to variations in the pipe's roughness which can differ from nominal values or non-ideal geometric factors of some of the fittings when compared to data. Systematic errors are also seen through the accuracy and sensitivity of the differential pressure gauges and the

precision of the inner diameters and measuring lengths which can generate into the calculated K and f values.

Quantitative Error Analysis is seen as their different sources of errors when determining the numerical uncertainty for f , K , and C_f . The main source of error is from the measured variables, the differential pressure, the temperature, the volumetric flowrate and the dimensions. There is also uncertainty in the pressure drop seen through the DPT fluctuations which impact the friction factor and flow coefficient equations. Another source of uncertainty is through the flow rate which was gotten through the rotameter as the team calibrated the rotameter in the pre-lab portions. Uncertainty in temperature measurement go to the calculations through the values of fluid density and viscosity which are needed to calculate various coefficients. Error propagation techniques need to be applied to the equations for the coefficients to evaluate the total uncertainty in the final reported values.

9. Appendix III: Sample Calculations

Trial 1- 40 SCFM

$$C_f = \frac{Q}{A\sqrt{2\Delta P/\rho}} \quad (Eq9)$$

$$= \frac{35.9148}{60(.1193)} \sqrt{\left(\frac{1000}{2(3.86)(6894.76)}\right)} = .69 @ 45 SCFM$$

Trial 1- Contraction 40 SCFM

$$f = \frac{\Delta PD}{2L\rho V^2} (EQ3) \quad \text{ } \frac{1103(.02)(10)}{2(.267)(997)(.45^2)} = .021$$

$$K = f \frac{L}{D} (EQ6)$$

Trial 1- 40 SCFM Contraction fitting

$$\text{Re} = \frac{\rho V D}{\mu} (EQ8) \quad \text{ } \frac{(997)(.45)(.026)}{.00653} = 1782$$

10. Appendix IV: Individual Team Contributions

NAME: Miguel

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	4	Operated lab equipment and calibrated measurements
Pre-Lab Editing		
Experimental Objective (Prelab)	0.4	Wrote the experimental objective
Apparatus (prelab)	1.4	I wrote the table and used image description for apparatus
Materials & Supplies (prelab)		
Procedure (prelab)		
Project Safety Evaluation (prelab)		
Final Lab Editing	1	
Executive Summary (final lab)		
Introduction (Final lab)		
Theory (Final Lab)		
Results (final lab)	8	Performed all requested calculations and

		calibrations
Discussion (final lab)		
References (final and/or prelab)	1	
Data Tabulation/Graphs (final)	.5	Input all data and graphs into
Error Analysis (final lab)		
Sample Calculations (final Lab)		
Power Point Presentation	2	Worked on Presentation
Total	18.3	

NAME: Christian

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	4	Operated apparatus
Pre-Lab Editing		
Experimental Objective (Prelab)		
Apparatus (prelab)		
Materials & Supplies (prelab)		
Procedure (prelab)	2	Wrote the Procedure
Project Safety Evaluation (prelab)		
Final Lab Editing	1	
Executive Summary (final lab)	1	Wrote the executive summary
Introduction (Final lab)		
Theory (Final Lab)		
Results (final lab)		
Discussion (final lab)	2.5	Wrote the discussion section
References (final and/or prelab)		
Data Tabulation/Graphs (final)		

Error Analysis (final lab)		
Sample Calculations (final Lab)	1	Wrote the sample calculations
Power Point Presentation	1	
Total	12.5	

NAME: Ify

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	4	Worked on the apparatus
Pre-Lab Editing		
Experimental Objective (Prelab)		
Apparatus (prelab)		
Materials & Supplies (prelab)	1	Wrote the materials and supplies
Procedure (prelab)		
Project Safety Evaluation (prelab)	1	Worked on the Project safety
Final Lab Editing	1	
Executive Summary (final lab)		
Introduction (Final lab)	2	Wrote the introduction
Theory (Final Lab)	2	Wrote the theory
Results (final lab)		
Discussion (final lab)		
References (final and/or prelab)		
Data Tabulation/Graphs (final)		
Error Analysis (final lab)	3	Wrote the error analysis
Sample Calculations (final Lab)		
Power Point Presentation	14	
Total		

